MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)
TECHNICAL SERVICES & OPERATIONS DIVISION

TECHNICAL MEMORANDUM & NOTE FOR APPROVAL (NFA)
INTER-OFFICE MEMO | REF: **MRPL/CDU-II/NFA/2026/014** | CLASSIFICATION: **CONFIDENTIAL**

TO:	Chief General Manager (Technical Services & Operations)
THROUGH:	General Manager (Maintenance & Reliability)
FROM:	Senior Manager (Mechanical Maintenance, CDU-II)
DATE:	05-MAR-2026
SUBJECT:	Approval for Emergency Procurement of Spool Piece 4"-HC-1012 & Overhaul of P-101B Seal Assembly

1. BACKGROUND & PROBLEM STATEMENT

During the ongoing baseline integrity assessment and pre-turnaround survey of the Crude Distillation Unit (CDU-II), comprehensive Non-Destructive Testing (NDT) was performed on Crude Feed Booster Pumps P-101A/B and associated piping. Ultrasonic gauging report MRPL-NDT-UT-2026-089 and vibration diagnostic report MRPL-RMC-VIB-2026-114 reveal two critical integrity threats requiring immediate management approval and procurement action prior to the monsoon turnaround.

2. KEY TECHNICAL FINDINGS

• **Wall Thinning (P-101A Suction Line 4"-HC-1012-A1A-H)**: Thickness at Elbow #2 extrados has degraded to **3.42 mm** (nominal 6.02 mm). The ASME B31.3 calculated retirement thickness is **3.20 mm**. With the current corrosion rate of 0.28 mm/year, remaining service life is calculated at only **9.4 months**.
• **Standby Pump P-101B Unavailability**: Elevated axial vibration (7.65 mm/s RMS) and continuous pressure decay in API Plan 53B barrier system indicate imminent seal failure and hydraulic recirculation. P-101B cannot serve as a reliable standby.

3. RISK ASSESSMENT & MITIGATION PROPOSAL

Operating beyond June 2026 without pipe replacement exposes the refinery to a catastrophic loss of containment (LOC) hazard of 145 m³/h hot hydrocarbon feed at 1.8 bar(g). The following actions are recommended for immediate approval:
1. **Fabrication & Replacement of 4" Spool**: Pre-fabricate replacement spool piece in ASTM A106 Gr. B with internal cladding.
2. **OEM Mechanical Seal Overhaul**: Mobilize OEM service engineer for P-101B cartridge seal rebuild and accumulator recharge.
3. **Estimated Outlay**: Total expenditure of INR 14.50 Lakhs against Operational Maintenance Budget (CDU-II OPEX).

4. FINANCIAL & PROCUREMENT APPROVAL MATRIX

Item	Scope / Description	Vendor / Contractor	Estimated Cost (INR)	Budget Code
1	4" Carbon Steel Spool (ASME B16.9 Elbows, cladded)	L&T Heavy Engineering / Approved Fabricator	■ 6,80,000	OPEX-CDU2-26-01
2	API Plan 53B Seal Cartridge Rebuild Kit	Flowserve India Pvt. Ltd.	■ 5,20,000	OPEX-CDU2-26-02
3	Site Radiography, NDT & Third Party EIL Inspection	Engineers India Ltd (TPI)	■ 2,50,000	OPEX-CDU2-26-03
TOTAL	Emergency Turnaround Remediation	—	■ 14,50,000	APPROVED CAP

5. SIGN-OFF & INTER-DEPARTMENTAL CONCURRENCE

Initiated By	Concurred By	Finance Concurrence	Approved By
R. K. Sharma Sr. Mgr (Mechanical)	D. A. Gowda GM (Maintenance)	S. Sengupta DGM (Finance)	P. K. Verma CGM (Operations)
[Signed 05/03/2026]	[Signed 05/03/2026]	[Concurred 06/03/2026]	[APPROVED 06/03/2026]